

Markscheme

Mathematics: analysis and approaches
Practice question bank

170. (a)

Three planes are given by $x + 2y - z = 3$, $2x + y + z = 5$, and $3x + 3y + kz = 8$.

Find the value of k for which the three planes do not meet at a unique point.

a unique point of intersection fails when the determinant of the coefficient matrix is zero (M1)

$$\det = 1(k - 3) - 2(2k - 3) - 1(6 - 3) \quad (M1) \text{ A1}$$

$$= -3k \quad \text{A1}$$

$$\text{set } -3k = 0: k = 0 \quad \text{A1}$$

[6 marks]